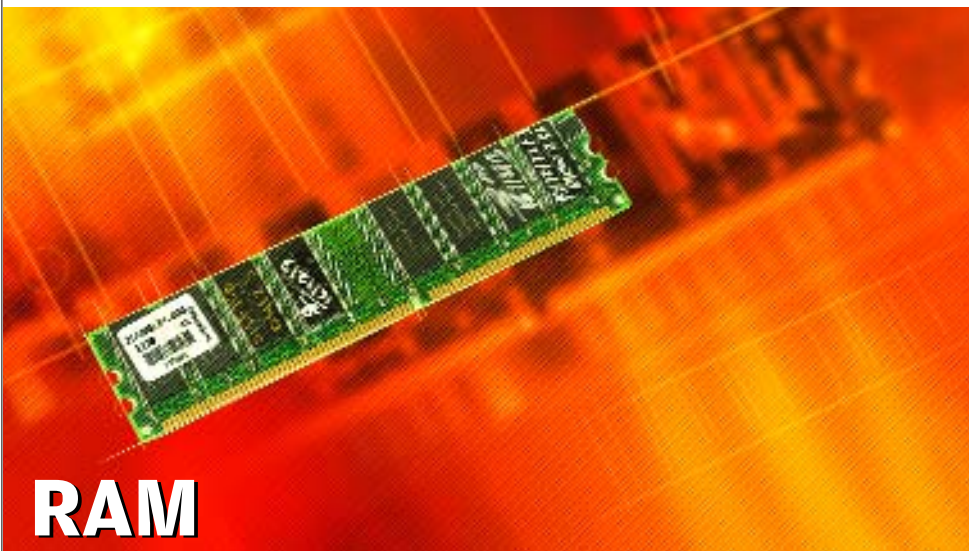




RAM

More often than not, the focus in RAM is on speed, and DDR2 running at as high as 667 MHz is already out there. The DIMM has not seen much change. DDR, however is using FBGA (Fine Ball Grid Array) instead of TSOP II (Thin Small-Outline Package). All in all, the DRAM module isn't scheduled for a massive change in the desktop segment.

However, the server segment witnesses an ever-increasing demand for RAM in a form factor that takes up very little physical space. Tape Carrier Packaging (TCP), Elevated Package Over CSP (EPOC) and Foldable Electronic Memory Module Assembly (FEMMA) are meant to address this problem.

The evolution of RAM modules from the slow EDO-RAMs to SDRAM, and now DDR-RAM, has always strived to push

the speed limit further. This will continue to be a trend in the future. With the advent of 64-bit processors that require ECC memory, the market will witness another major change. These ECC memory modules, which were only used in servers thus far, will eventually trickle down to the desktop arena.

RAM Demystified

Address Line: This carries either a 0 or a 1 signal to the transistors. There are two address lines for a series of transistors. If one address line is carrying a "1" signal, the other address line carries a "0" signal.

Data line: This carries the cumulative data signal from the RAM to the system bus.

Capacitor: This holds the information—either a 0 or a 1—depending on whether it is in a charged or a discharged state.



Buying Tips

- ✓ **MINIMUM RAM:** Most operating systems such as Microsoft Windows XP need at least 256 MB of RAM to run as intended. Add to that the requirements of applications, and you'll see that 512 MB of RAM is the ideal amount to get.
- ✓ **DDR OR SD RAM:** Given a choice, always opt for DDR memory. Performance is superior, and the SD variant is now almost obsolete.
- ✓ **FREQUENCY:** The frequency of the RAM has to match the one supported by the motherboard. Buy the correct speed RAM as specified in your motherboard manual—it depends on the chipset the motherboard uses.
- ✓ **SINGLE OR MULTIPLE:** As far as possible, use a single RAM module. The advantages are manifold. First, a single stick costs less than multiple

modules put together. Secondly, there won't be any compatibility issues, and you will still have a vacant slot for a further upgrade.

- ✓ If you are buying a new motherboard that's based on the latest chipsets (nForce3, 915, KT800 etc.), opt for 400 MHz memory modules, which are currently the fastest available.
- ✓ If you are buying two RAM modules for gaining the dual-channel advantage, make sure they are matched pairs—that is, that their frequency, timing and other parameters are matched by the memory manufacturer
- ✓ Always buy branded RAM modules sold in proper packaging. Lifetime warranties (generally extending to a 5-year period) are also offered, albeit at a price.

Ready Reckoner



You Need

To upgrade RAM on an olderboard that has no support for DDR-RAM

Look for

SDRAM that preferably has a frequency of 133 MHz



You Need

A cost-effective solution

Look for

DDR-RAM that has a frequency in the range of 266 MHz to 333 MHz



You Need

A performer—cost no bar

Look for

400 MHz DDR-RAM



Do Remember

1 Enable the lowest CAS setting for your RAM in the BIOS. Also, set it to use the highest bus speed setting it supports.

2 Check the RAM modules for compatibility with your motherboard. The list of validated memory modules can be found on the motherboard manufacturers Web site.

3 Always buy genuine memory modules to avoid compatibility issues. These have holograms from their respective manufacturers on them.

4 Consider buying low latency memory modules, if you intend to over-clock the memory modules for your gaming rig.

5 When buying costly memory modules such as Kingston, Corsair etc, make sure they are covered by life-time warranty.

6 Presently there are different memory modules available for Pentium III and P4 motherboards, hence make sure you buy the right one as you upgrade.

7 Memory modules with heat spreader should be considered when building high end rig, since they help in dissipating the heat.